

Executive Summary

Iris Classification using Machine Learning

Intern Information

Name: Tushar Patel

Internship: Machine Learning Internship

Organization: Alfido Tech

Project: Iris Classification using Machine Learning

Project Objective

The objective of this project was to develop a machine learning classification model capable of predicting the species of an Iris flower based on its physical characteristics. The model utilizes four botanical measurements—Sepal Length, Sepal Width, Petal Length, and Petal Width—to classify flowers into one of three species: Iris-setosa, Iris-versicolor, or Iris-virginica.

Dataset Overview

The project was implemented using the Iris Classification Dataset consisting of 150 flower samples equally distributed among three species. The dataset contains four numerical features and one target variable representing the flower species.

Dataset Statistics

Attribute	Value
-----------	-------

Total Records	150
---------------	-----

Features	4
----------	---

Classes	3
---------	---

Missing Values	0
----------------	---

The dataset was balanced and required no additional preprocessing for missing data handling.

Methodology

A structured machine learning workflow was followed:

- Data Loading and Exploration
- Exploratory Data Analysis (EDA)
- Data Visualization

4. Feature Selection
5. Train-Test Split (80:20)
6. Model Training
7. Model Evaluation
8. Model Saving and Inference

Visualization techniques such as scatter plots and pair plots were used to understand feature relationships and class separability. Analysis revealed that Petal Length and Petal Width were the most influential features for classification.

Machine Learning Models Implemented

The following classification algorithms were trained and evaluated:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Decision Tree Classifier

Model performance was assessed using Accuracy, Confusion Matrix, Precision, Recall, F1-Score, and Classification Report.

Results and Performance

Model	Accuracy
Logistic Regression	100%
K-Nearest Neighbors	100%
Decision Tree	100%

All three models achieved perfect classification performance on the test dataset. Logistic Regression was selected as the final model due to its simplicity, efficiency, interpretability, and excellent predictive performance.

Key Findings

- The dataset was clean and balanced with no missing values.
- Petal-based features showed the strongest predictive capability.
- Iris-setosa was clearly separable from the other species.
- All implemented algorithms achieved 100% classification accuracy.

- Logistic Regression was selected as the final deployment model.
-

Conclusion

This project successfully demonstrated the complete machine learning pipeline, including data exploration, visualization, model development, evaluation, and model persistence. The final Logistic Regression model was saved using Joblib and validated through inference testing. The project provided practical experience in supervised machine learning and established a strong foundation for future classification and predictive analytics projects.

Project Repository

GitHub Repository: <https://github.com/BinaryTp/iris-classification-ml>